3.7 Refrigerators

Refrigeration is the process of continuous cooling or extraction of heat to below that of the atmosphere from a substance with a help of the external work. It is based on the second law of thermodynamics that the heat can made to flow from cold body to a hot body with the help of external source.

Refrigerator is a machine which produces cold. It is used to remove heat from the refrigerated space and reject it to atmosphere. Hence it maintains the temperature below the surrounding atmosphere. The working fluid used in refrigerator is called refrigerant. Ex: Ammonia, Methyl chloride, Freon etc.

Parts of a Refrigerator and their functions

Evaporator: It is used to receive the liquid refrigerants from the condenser through throttle valve and to evaporate it. It is also called as cooling unit or freezing unit.

Compressor: It is used to draw the refrigerant from the evaporator through suction valve at low pressure and to force it to the condenser through exhaust valve at high pressure and temperature.

Condenser: It is used to condense the refrigerants at saturation temperature from vapour

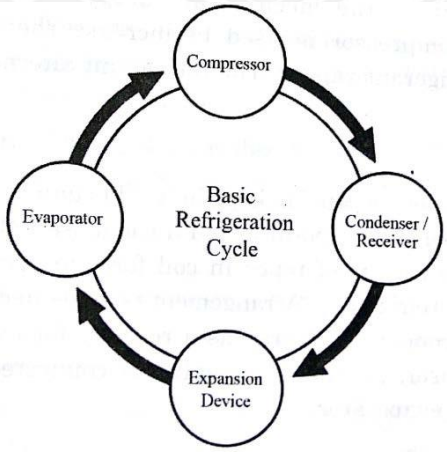


Fig 3.7.1- Condenser

to liquid by transferring its latent heat to the water.

Throttle (or) Expansion valve: It is used to control the rate of admission of refrigerants to the evaporator. As the refrigerant expands in the valve, it loses its pressure and temperature at its exit end and gets cooled.

Absorber (or) Receiver: It is a unit contains water. It is used to receive the refrigerant

from the evaporator and to absorb it with water to become a strong solution of ammonia with an increase in temperature.

Pump: It is to pump the refrigerant from the absorber to the separator.

Separator: It is used to separate the refrigerant vapour from the strong solution using heat energy and then send it to condenser.

Based on the external work it mainly classified in to two types.

1. Using mechanical energy – vapour Compression Refrigeration
2. Using heat energy – vapour Absorption Refrigeration

Vapour Compression Refrigeration Principle

In this technique, the mechanical energy is utilized to achieve refrigeration. A compressor is used to increase the temperature and pressure of the refrigerant vapour. The refrigerant alternatively evaporates and condenses.

Construction

It consists of an evaporator is kept in a cold chamber. A compressor is connected in between an evaporator and a condenser. Both the condenser and the evaporator consist of pipes in coil form to provide more contact surface area for the refrigerant. Arrangement is made to circulate cold water or air around the condenser. A tray as a receiver for exit water is placed nearby the condenser. An expansion valve is connected in between the condenser and the.

Working

The unit completes one cycle with the following operations such as Suction, Compression, Condensation, Expansion and Vaporization.

Suction: The low-pressure vapour in dry state is drawn from the evaporator during in suction of the compressor through its inlet valve.

Compression: The pressure and temperature of the refrigerants increases inside the compressor and it is discharged through the exhaust valve and enters the condenser.

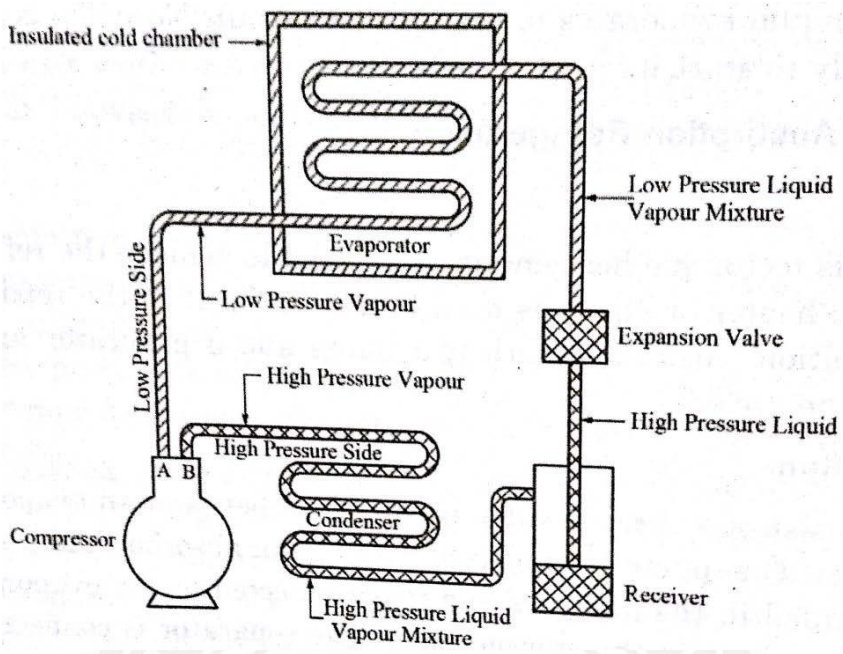


Fig 3.7.2- Refrigeration

Condensation: The condenser absorbs its heat. The heat now flows from condenser to the cooling water, which is circulated around it. As a result vapour is cooled and a return to liquid state. It is then made to exit through the expansion valve.

Expansion: The exit or expansion of the liquid result in decrease of its temperature and pressure to about -10^0C and enters the evaporator kept in the storage room.

Vaporization: As the temperature of the liquid is less than that of the storage room, the heat will be absorbed by it from the room. As the result the liquid evaporates to almost dry vapour. Now this is dry vapour and is ready to start its next cycle.

Vapour Absorption Refrigeration Principle

In this technique heat, energy is utilized to achieve the refrigeration. An electric heater or steam is used to add the heat to the refrigerant for its evaporation. In addition, an absorber, a pump and a generator are used to complete the cycle.

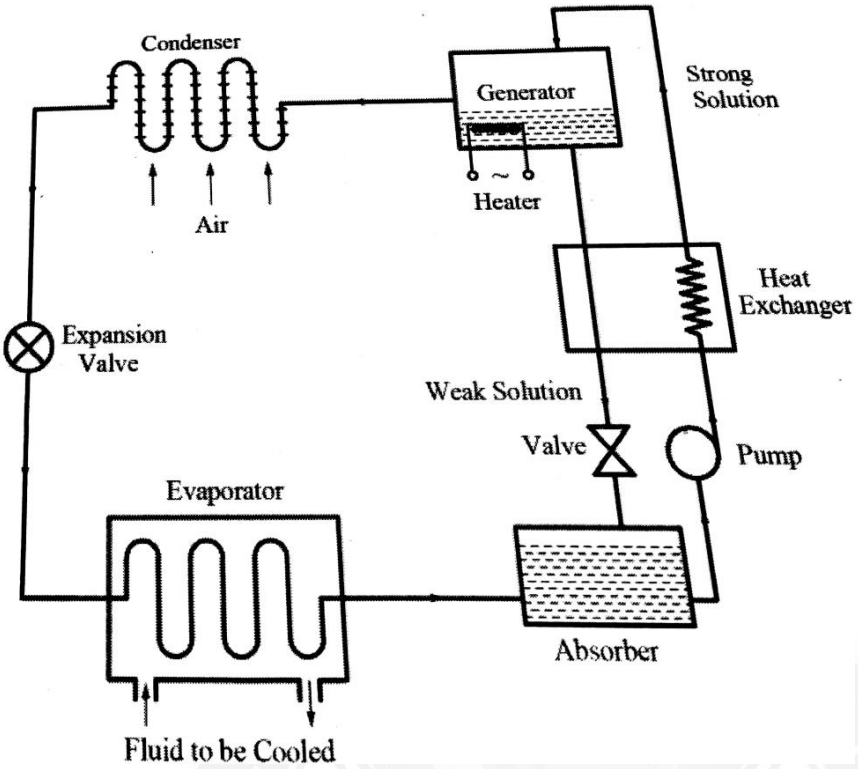


Fig 3.7.3- Refrigeration principle

Construction

It consists of a throttle valve is connected between an evaporator and a condenser. One pump is connected between an absorber and a separator. Water is filled in the absorber and it is connected to the evaporator. The evaporator is kept in the storage room. The separator is connected to the condenser. Both the condenser and evaporator consist of pipes in coil from to provide more contact surface area for the refrigerant. Arguments are made to circulate the cold water around the condenser and in the absorber. An electric heater is housed in the separator. Trays may be positioned to collect the exit water near by the absorber and condenser. A receiver is connected in between the condenser and the throttle valve.

Working

Dry ammonia vapour from the evaporator enters the absorber containing water it is absorbed by the water becomes a strong ammonia solution with an increase in temperature. The heat generated during this process is removed to some extent by circulating cold water through a pipe. Otherwise absorbing capacity reduces with hot water.

A pump to the generator now pumps the strong ammonia solution where it is heated by an electric coil. As result, ammonia vaporizes and separates out from the water. It is then driven out from the solution to the condenser where it is condensed and return to the liquid state.

The liquid ammonia is then collected in the receiver. The high-pressure liquid ammonia is

then passed through the throttle valve where it is expanded with decrease in temperature and pressure. Later it enters the evaporator kept in the storage room.

Comparison between vapour compression and vapour absorption refrigeration

Vapour compression refrigeration	**Vapour absorption refrigeration**
Works using mechanical energy	Works using heat energy
Refrigeration capacity is less than 1000 tons.	Refrigeration capacity is greater than 1000 tons.
COP is much higher	COP is less.
Noisy due to compressor.	Pump noise is less.
Chances of leakage of refrigerant are more.	No leakage.
Maintenance and operating cost are high.	Less.
Smaller in size.	Larger
Wear and tear are more.	Wear and tear are less

Properties of good (ideal) refrigerants

An ideal refrigerant should have the following properties

1. Low viscosity
2. Low freezing point
3. Low boiling point
4. Low heat capacity
5. Low specific volume
6. Low saturation pressure
7. Odourless
8. High latent heat of vaporization
9. Good thermal conductivity.
10. High COP (Coefficient of performance = Output/input = Q/W)
11. Non inflammable and non-explosive.

12. High critical pressure and temperature.

Types of refrigerants

1. **Ammonia:** Ice plants, large cold storages, etc
2. **Carbon dioxide:** Marine refrigerators, dry ice making and air conditioning.
3. **Methyl chloride:** domestic and industrial refrigerators
4. **Fluorocarbon refrigerants:** Freon-11, 12, 13, 22, 113, 114, etc.

   These are used in domestic refrigerators, water coolers, air conditioning plants cold storages etc.
5. **Sulphur dioxide:** Domestic refrigerators

Uses of refrigeration

1. It is used make ice
2. It used to preserve food products and medicines.
3. It used in Air conditioning.
4. It used as a rocket fuels.
5. It used in transportation of foodstuffs, dairy products.
6. It used in special industrial processes.